# App Dev I - Project Report

## 1. Student Details

**Name:** Aditi
**Roll Number:** DS25F2006635
**Email:** 25f2006635@ds.study.iitm.ac.in
**About Me:** I am a student pursuing the IIT Madras BS Degree program along with my undergraduate studies in Physics. I have a keen interest in technology, data science, web application development, and building practical applications that solve real-world problems.

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## 2. Project Details

### Project Title: Trekkmaster – Trekking Management Application

### Problem Statement:

To design and develop a web-based **Trekking Management Application** that simplifies the management of trekking activities for adventure organizations.

Traditional trekking organizations often rely on spreadsheets, phone calls, and manual coordination for managing treks, staff, participants, and bookings. This can lead to difficulties in managing trek availability, assigning staff, tracking bookings, preventing overbooking, and maintaining trekking history.

The application provides separate role-based interfaces for **Admin, Trek Staff, and Trekkers (Users)**, allowing each role to perform the operations relevant to them.

### Approach:

The application is developed using **Flask** as the backend framework, with **Jinja2, HTML and CSS** used for the frontend and **SQLite** used as the database.

The application follows a role-based architecture consisting of three major user roles:

- **Admin** – manages treks, users, trek staff, and bookings.
- **Trek Staff** – manages assigned treks and participants.
- **Trekker/User** – explores available treks, makes bookings, manages bookings, and views trekking history.

The database is created programmatically and contains the required entities and relationships for managing users, staff, treks, and bookings.

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# 3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** to understand the project requirements, planning the application structure and designing database relationships and ER diagrams

The extent of AI/LLM usage is around **8-10%**, limited to **code suggestions and documentation formatting**.

All final implementation logic, debugging, and integration were done manually.

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# 4. Technologies and Frameworks Used

| Technology / Library | Purpose |
|---|---|
| **Flask** | Core backend web framework |
| **SQLAlchemy** | Object Relational Mapper for SQLite database |
| **Jinja2** | Template engine for rendering dynamic HTML pages |
| **Flask-Login** | User authentication and session management |
| **SQLite** | Lightweight local database for storing user data |

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# 5. Database Schema / ER Diagram

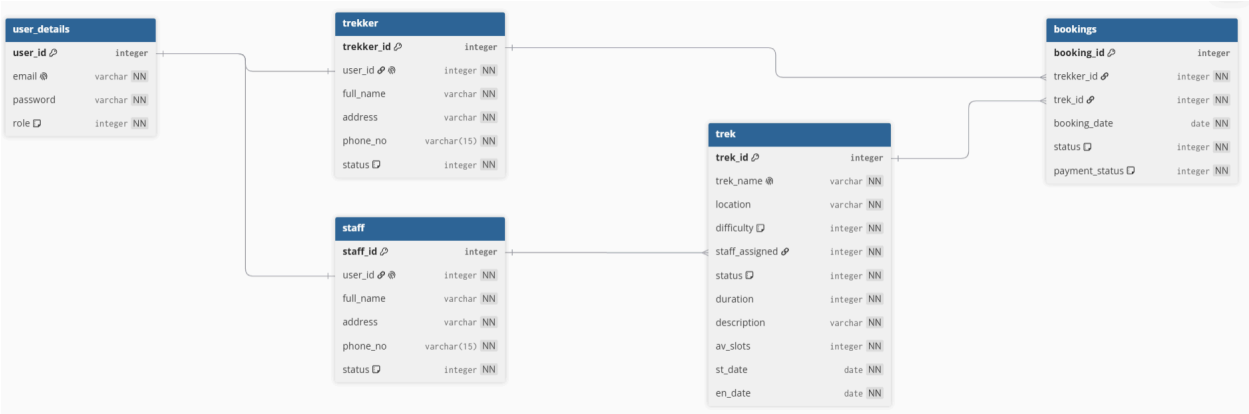
**Tables:**

1. **User_Details** — stores authentication credentials and system role details (user_id, email, password, role)
2. **Trekker** — stores profile information for trekking participants (trekker_id, user_id, full_name, address, phone_no, status)
3. **Staff** — stores profile details for assigned staff members (staff_id, user_id, full_name, address, phone_no, status)
4. **Trek** — stores trekking event information and schedule details (trek_id, trek_name, location, difficulty, staff_assigned, status, duration, description, av_slots, st_date, en_date)

5. **Bookings** — tracks trek bookings and payment statuses (booking_id, trekker_id, trek_id, booking_date, status, payment_status)

**Relationships:**

- One-to-One → `User_Details → Trekker`
- One-to-One → `User_Details → Staff`
- One-to-Many → `Staff → Trek`
- One-to-Many → `Trekker → Bookings`
- One-to-Many → `Trek → Bookings`




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# 6. Video Presentation

**Drive Link:**

Project Video.mp4 *(Accessible to all with "View" permission.)*